Akoizo Original Practice - Form 1 - Answer key

Division B | Roller Coaster | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Which energy source is allowed to move the ball besides its own gravity?

A. Simple machines in the elevators

B. A wound spring

C. A rubber band

D. A small battery motor

Correct: A

2. (2 points) A device measures 50.0 x 22.0 x 70.0 cm. What is its Size Score?

A. 18

B. 38

C. 28

D. 142

Correct: C

3. (2 points) Why does a second-class lever need help to make IMA less than 1?

A. It is not a lever

B. Its IMA is always 1

C. On its own, its IMA is greater than 1

D. It cannot move

Correct: C

4. (2 points) Which change reduces energy losses?

Unit: Energy, friction, and the rolling ball.
Worked-case data: A solid ball drops 0.40 m from rest without losses. Find its speed.
Separate investigation observations:
Predicted speed: 2.4 m/s at the bottom of a 0.40 m drop.
Measured speed: 2.0 m/s.
Track: Three joints along the drop.

A. Using a softer track

B. Making curves tighter

C. Smoothing joints between track pieces

D. Adding more bumps

Correct: C

5. (2 points) Which run counts for the final score?

Unit: Testing and competition day.
Worked-case data: Target 52 s. Practice run: 50.6 s. The calibration table says each screw turn adds 0.4 s. What should the team do?
Separate investigation observations:
Test week 1: Loop failed 3 of 10 runs.
Cause: Ball hit a raised joint just before the loop.
After sanding joint: Loop worked 10 of 10.

A. The better of the two runs

B. The first run

C. The last run

D. The average

Correct: A

6. (4 points) Describe one fine and one coarse time adjustment.

Unit: Hitting the Target Time.
Worked-case data: Calibration: base path 34 s; each slope-screw turn adds 0.8 s; removable spiral adds 9 s. Plan for a Target Time of 46 s.
Separate investigation observations:
Runs at setting A: 41.8, 42.1, 41.9 s.
Runs at setting B: 44.6, 45.4, 44.8 s.
Target Time: 45 s.

answer: Fine: a screw that changes the slope of a long section. Coarse: adding or removing a spiral or switchback. (4 points).

7. (4 points) What labels must a loop have?

Unit: Loops: physics and scoring.
Worked-case data: A loop has a 9.0 cm path radius. What is the ideal minimum release height above the loop bottom, and a practical target?
Separate investigation observations:
Release 24 cm: Ball fell from top of loop in 6 of 10 trials.
Release 28 cm: Ball completed loop 10 of 10.
Loop path radius: 9 cm.

answer: A clearly labeled point at the POI and a labeled measurement from the POI to the highest inside point. (4 points).

8. (4 points) Calculate the maximum Time Score for a Target Time of 40 s.

answer: 40 x 5 = 200 points. (4 points).

9. (4 points) Why recheck calibration after transport?

Unit: Size and construction.
Worked-case data: Option 1: 55 x 25 x 75 cm with a 22 cm loop. Option 2: 45 x 20 x 65 cm with a 15 cm loop. Compare Size + Loop points.
Separate investigation observations:
Home runs: Consistent 44.7 s.
After car ride: Runs 42.1 s; base slightly twisted.
After leveling: Runs 44.6 s.

answer: Small shifts in tilt or slope change run times, so the device may no longer hit its calibrated times. (4 points).

10. (4 points) How can elevators help with a long Target Time?

Unit: Elevators: levers, wheels, and pulleys.
Worked-case data: A second-class lever has IMA 1.6. It drives a movable pulley inverted to give IMA 0.5. Is the combination legal, and how far does the ball rise if the effort moves 8 cm?
Separate investigation observations:
Elevator test: Counterweight drops 5 cm; ball rises 9 cm.
Requirement: Ball must rise at least 10 cm.
Team claim: Elevator earns 50 points.

answer: Lifting the ball takes time and gives it new height to spend on slow track sections. (4 points).

11. (6 points) Solve this case and explain your reasoning: A solid ball drops 0.40 m from rest without losses. Find its speed.

reasoning: Energy: mgh = (7/10)mv^2.
v = sqrt (10gh ÷ 7) = sqrt (10 x 9.8 x 0.40 ÷ 7) = sqrt 5.6.
v about 2.37 m/s. (3 points).

conclusion: About 2.4 m/s; a sliding object would reach about 2.8 m/s. (3 points).

12. (2 points) What does this show?

Supplied synthetic observations:
Test week 1: Loop failed 3 of 10 runs.
Cause: Ball hit a raised joint just before the loop.
After sanding joint: Loop worked 10 of 10.

A. The ball was too heavy

B. Fixing one recorded failure cause made the loop reliable

C. Taller loops are always better

Correct: B

13. (2 points) Which setting should the team use?

Supplied synthetic observations:
Runs at setting A: 41.8, 42.1, 41.9 s.
Runs at setting B: 44.6, 45.4, 44.8 s.
Target Time: 45 s.

A. Setting B with a small increase

B. Setting A

C. Setting B with a small decrease

Correct: A

14. (2 points) Define “Centripetal force” in the context of Roller Coaster.

answer: Net inward force for circular motion. (2 points).

15. (2 points) Define “Size Score” in the context of Roller Coaster.

answer: (60 - H) + (30 - W) + (80 - L). (2 points).

16. (2 points) Define “Layered track” in the context of Roller Coaster.

answer: Track stacked at different heights. (2 points).

17. (2 points) Define “Type 1 elevator” in the context of Roller Coaster.

answer: First-class lever + wheel and axle, IMA < 1. (2 points).

18. (2 points) Define “Height budget” in the context of Roller Coaster.

answer: Plan for how starting height is spent. (2 points).